



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.4

Surface Area Using Nets

Lesson 10-3 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a solid by using its net.

Language Objective: I can explain my work using the words surface area, net, face, and two-dimensional.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Net

Face

Two-dimensional

Edge



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The total area of all the faces of a solid is its .

2

A flat pattern that folds up into a solid is a .

3

A flat surface of a solid figure is a .

4

A flat shape with length and width but no thickness is .

5

A line segment where two faces meet is an .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

For the box $l = 8$, $w = 5$, $h = 3$, you found three face pairs. How did you use the net to organize the faces into pairs, and which pair had the largest area?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Surface area** — The total area of all the flat sides of a solid.
Área de superficie — El área total de todos los lados planos de un sólido.

☐ **Net** — A flat shape that folds up into a solid.
Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.

☐ **Face** — One flat side of a solid shape.
Cara — Un lado plano de una figura sólida.

☐ **Two-dimensional** — A flat shape with length and width, but no thickness.
Bidimensional — Una figura plana con largo y ancho, pero sin grosor.

☐ **Edge** — The line where two flat sides of a solid meet.
Arista — La línea donde se unen dos lados planos de un sólido.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The faces come in ____ pairs because opposite faces are ____.

Las caras vienen en ____ pares porque las caras opuestas son ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A net lays every face flat so you can see and measure all six faces.

Evidence: Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

Reasoning: This evidence connects the definition of surface area to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The faces come in ____ pairs because opposite faces are ____.**
Las caras vienen en ____ pares porque las caras opuestas son ____.
- **The largest pair is ____ × ____ = ____, counted twice for ____ in².**
El par más grande es ____ × ____ = ____, contado dos veces para ____ in².

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in $\times$ 4 in $\times$ 5 in, a student might add just one face from each pair: $(9 \times 4) + (9 \times 5) + (4 \times 5) = 36 + 45 + 20 = 101 \text{ in}^2$, but the net actually has two of each face, so the correct total is $2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 \text{ in}^2$ — exactly double. Before you submit, ask: 'Did I count both matching faces in each pair, or only one from each pair?'*
7. **Try It 1:** A. 6 — *A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.*
8. **Try It 2:** A. 52 in^2 — *From the net: $SA = 2(2 \times 3) + 2(2 \times 4) + 2(3 \times 4) = 12 + 16 + 24 = 52 \text{ in}^2$.*